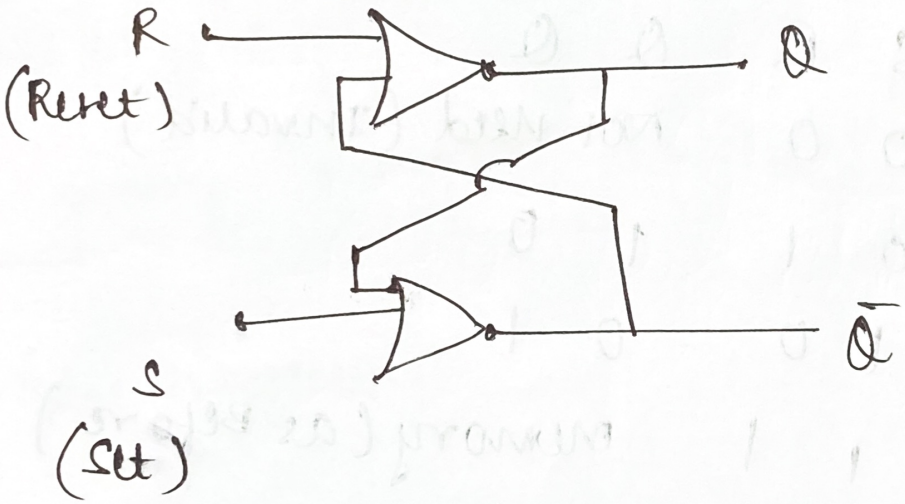
~~Sequential~~ Sequential circuits

Present output depends on present and past output / outputs

SR Latch

Basic storage element is called LATCH



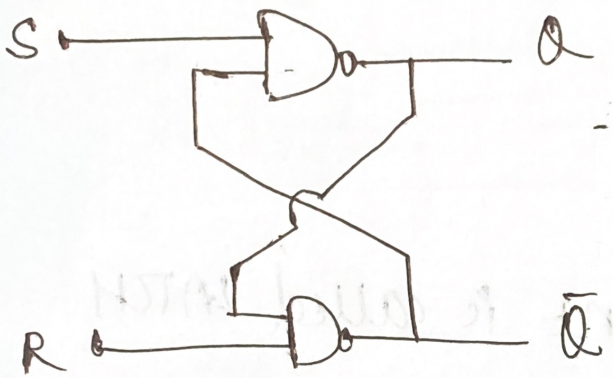
If Reset high then $Q = 0$

If set high then $Q = 1$

can be proved using truth table of NOR gate

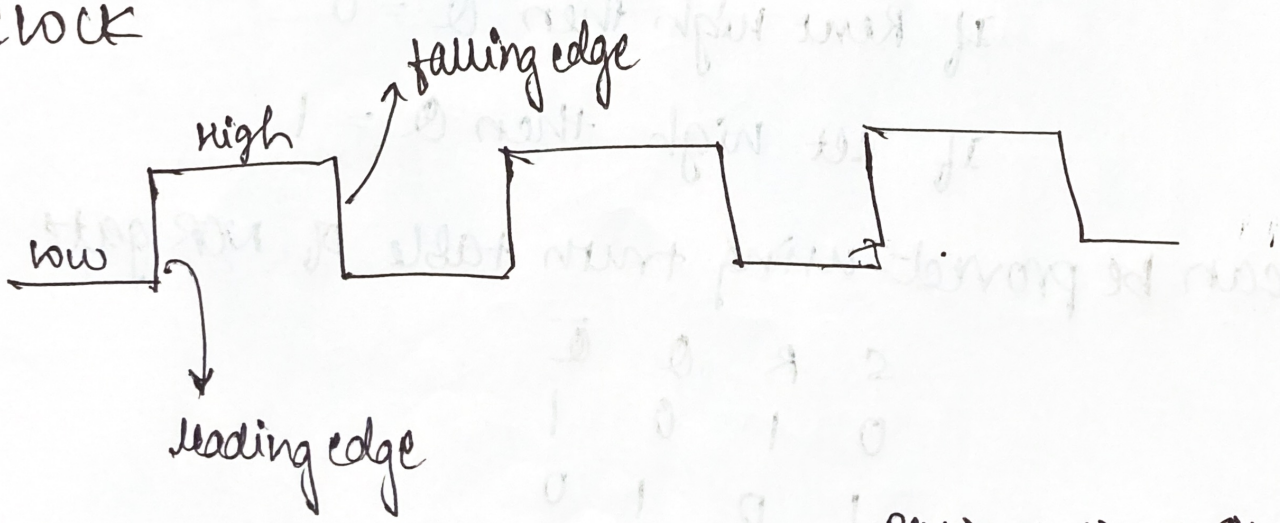
S	R	Q	$\bar{Q}$	
0	1	0	1	
1	0	1	0	
0	0	~~invalid~~		memory
1	1	~~no change~~		invalid

NAND SR Latch

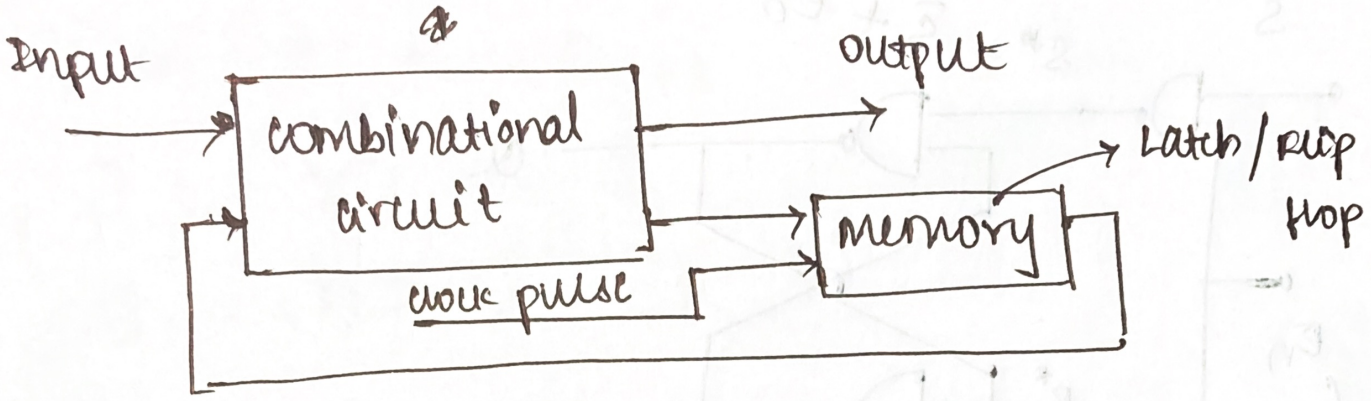


S	R	Q	$\bar{Q}$
0	0	Not used (Invalid)	
0	1	1	0
1	0	0	1
1	1	memory (as Before)	

CLOCK



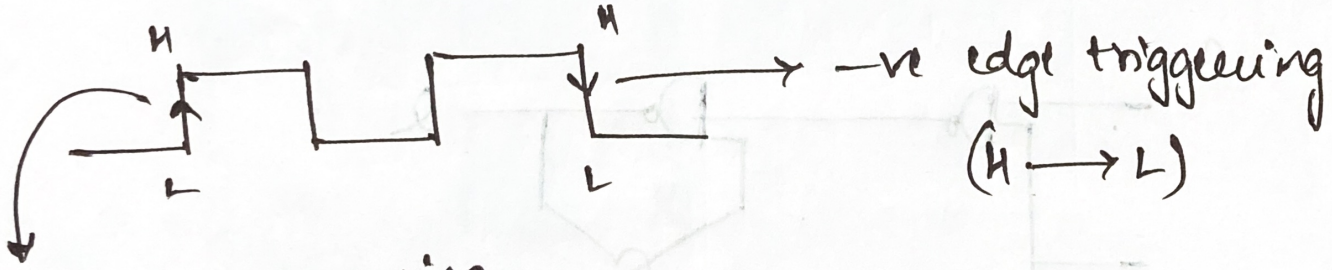
$$\text{Duty cycle} = \frac{\text{Ratio of time signal is high}}{\text{Total time}}$$



clock pulse change triggers change in memory

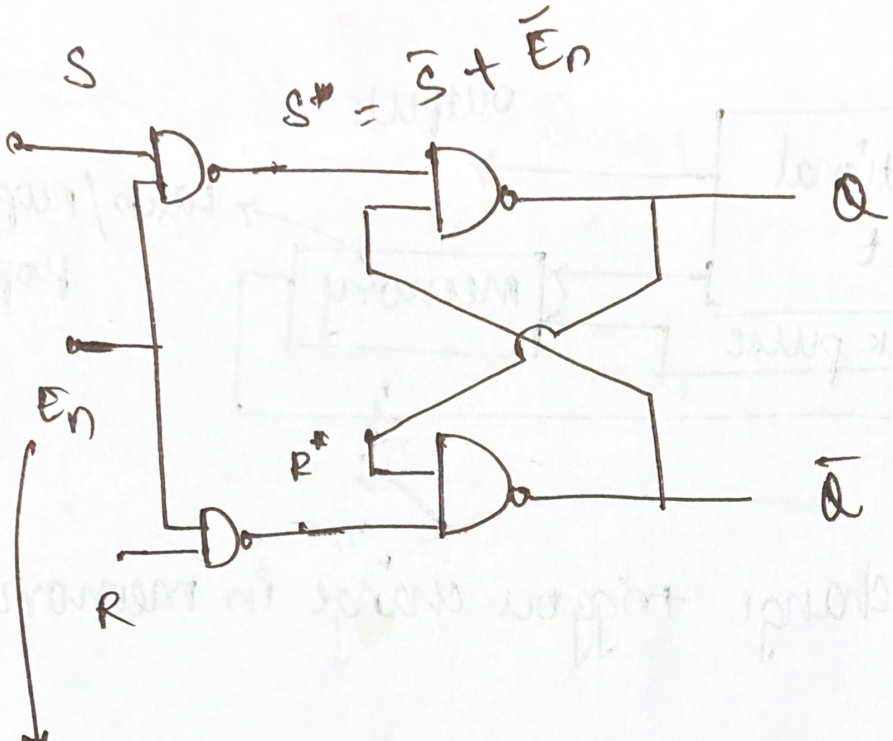
Level triggering

Edge triggering



+ve edge triggering

(L → H)

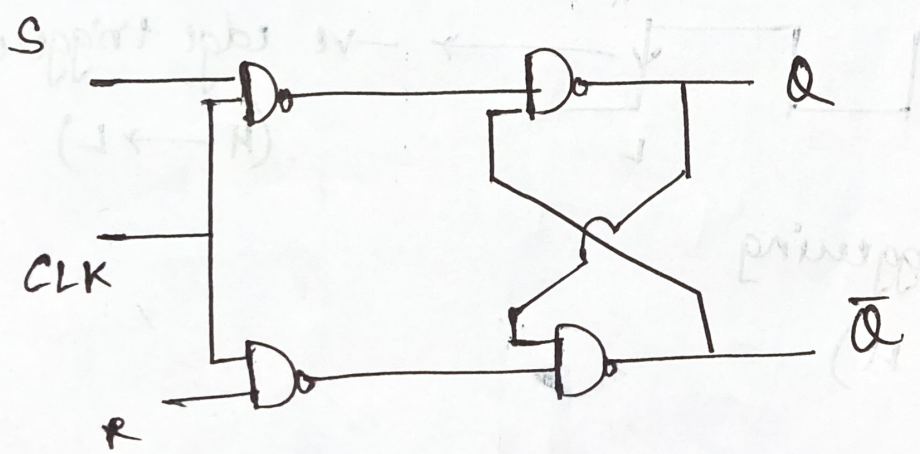


level triggered

will work only when $E_n = 1$

will act as latch

SR Flip Flop



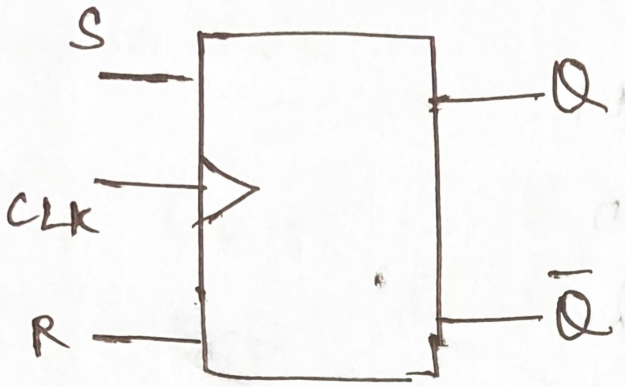
will work when clock goes H → L OR L → H

Latch is level sensitive.

CLK	S	R	Q_{n+1}
0	X	X	Q_n
1	0	0	Q_n
1	0	1	0
1	1	0	1
1	1	1	Invalid

Charaltoristic Table

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X



$$S^* = \overline{(S \cdot CLK)}$$

$$= \bar{S} + \overline{CLK}$$

$$R^* = \overline{(R \cdot CLK)}$$

$$= \bar{R} + \overline{CLK}$$

CLK	S	R	Q	$\bar{Q}$
0	X	X	Memory	
1	0	0	Memory	
1	0	1	0	1
1	1	0	1	0
1	1	1	Not used (Invalid)	

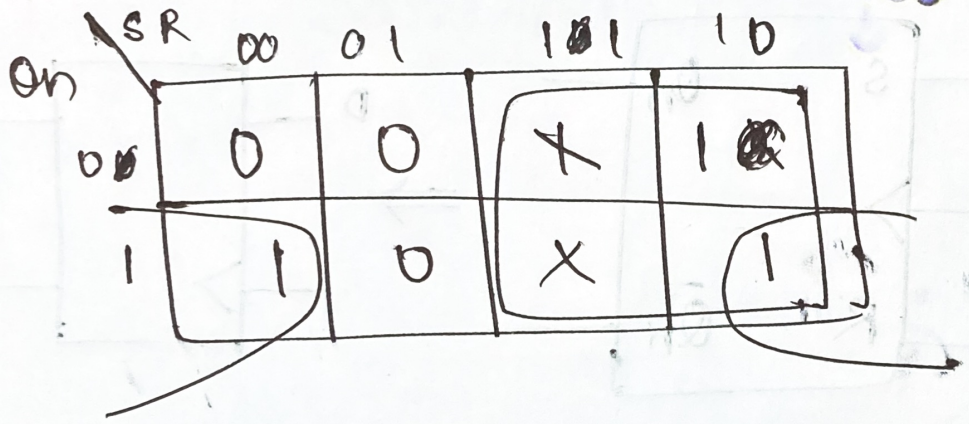
Q_{n+1} → Next state

Q_n → present state

Excitation Table

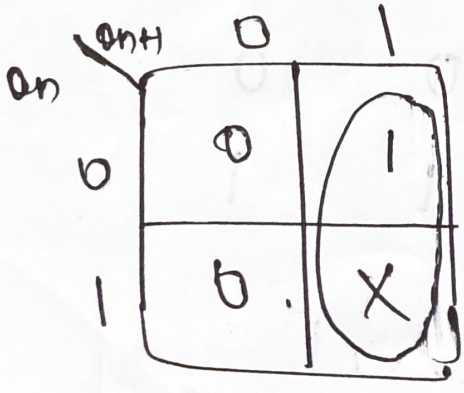
Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

From characteristic Table

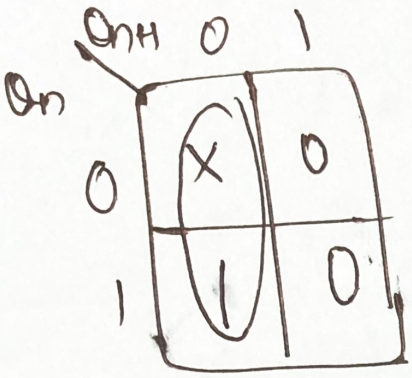


$$Q_{n+1} = S + Q_n\bar{R}$$

from Excitation Table

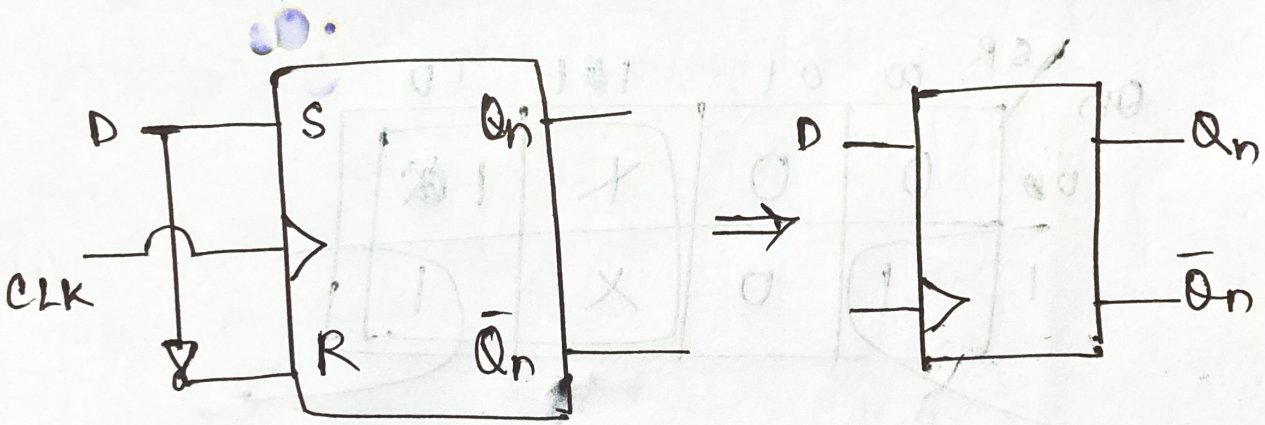


$$S = Q_{n+1}$$



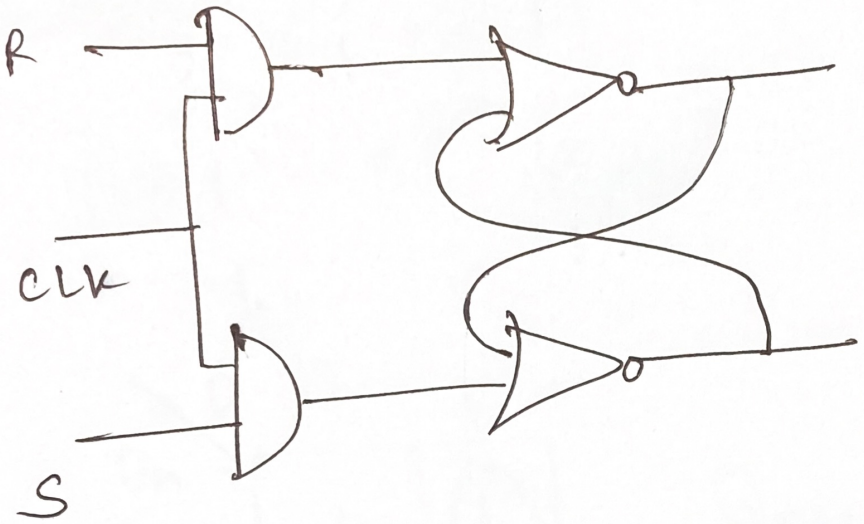
$$R = \overline{Q_{n+1}}$$

D Flip Flop (Data flipflop)



CLK	D	Q_{n+1}
0	X	Q_n
1	0	0
1	1	1

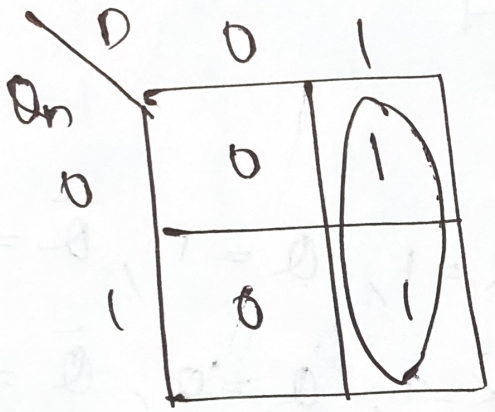
SR Flip flop implemented using AND NOR



Same truth table

Characteristic table

Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

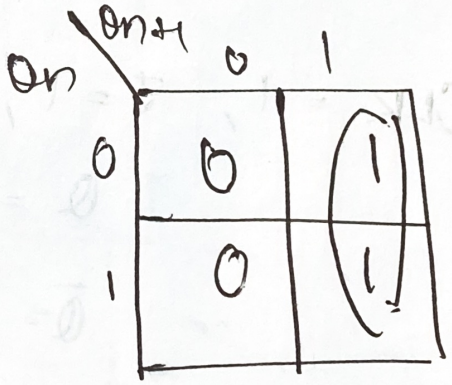


$$Q_{n+1} = D$$

Characteristic Equation

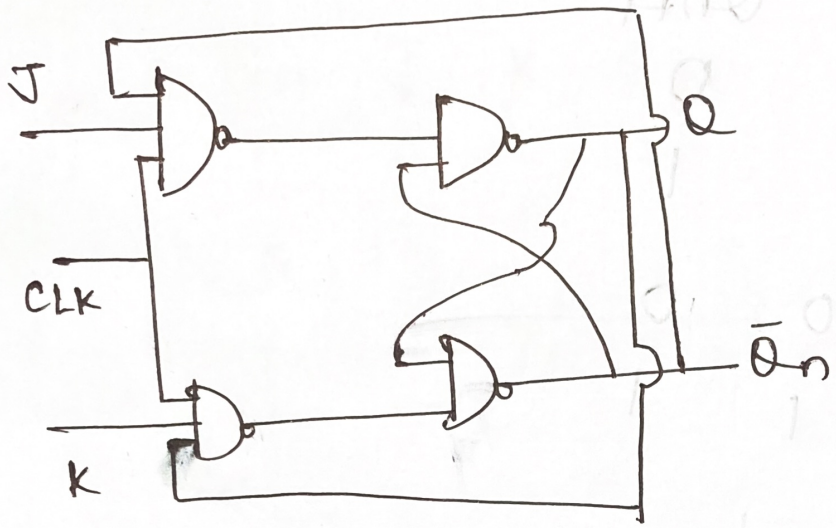
Excitation table

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1



$$D = Q_{n+1}$$

JK Flip Flop



CLK = 0, memory

CLK = 1, J = 1, K = 1, Q = 1, $\bar{Q}$ = 0

CLK = 1, J = 0, K = 1, Q = 0, $\bar{Q}$ = 1

CLK = 1, J = 1, K = 1

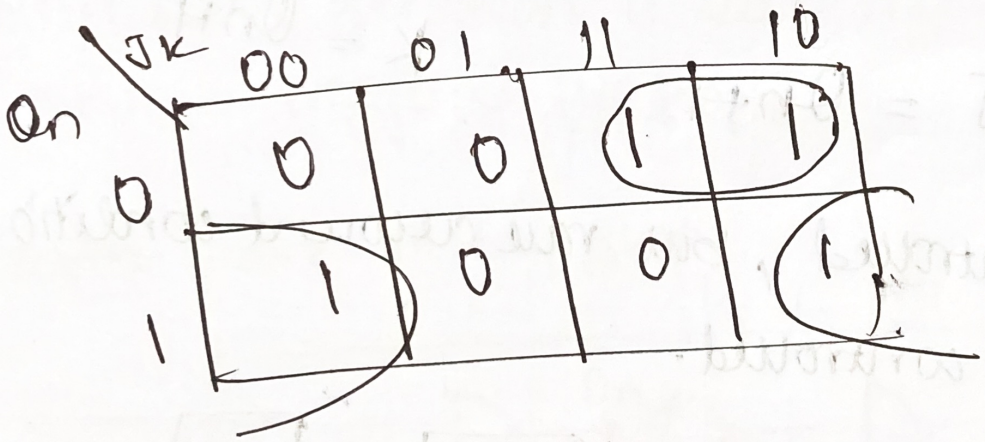
Q = 0, 1, 0, 1, 0

$\bar{Q}$ = 1, 0, 1, 0, 1

CLK	S	R	Q_{n+1}
0	X	X	Q_n (memory)
1	0	0	Q_n (memory)
1	0	1	0
1	1	0	1
1	1	1	$\bar{Q}_n$ (toggle)

characteristic Table

Q_n	J	K	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

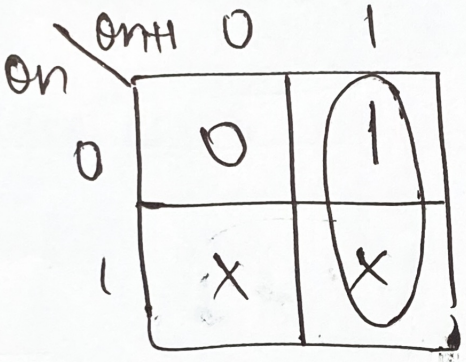


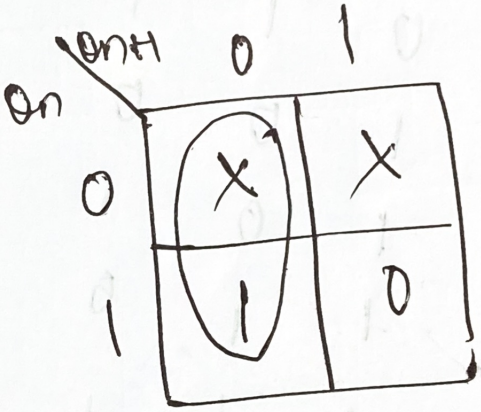
$$Q_{n+1} = \bar{Q}_n J + \bar{K} Q_n$$

characteristic Equation

Excitation Table

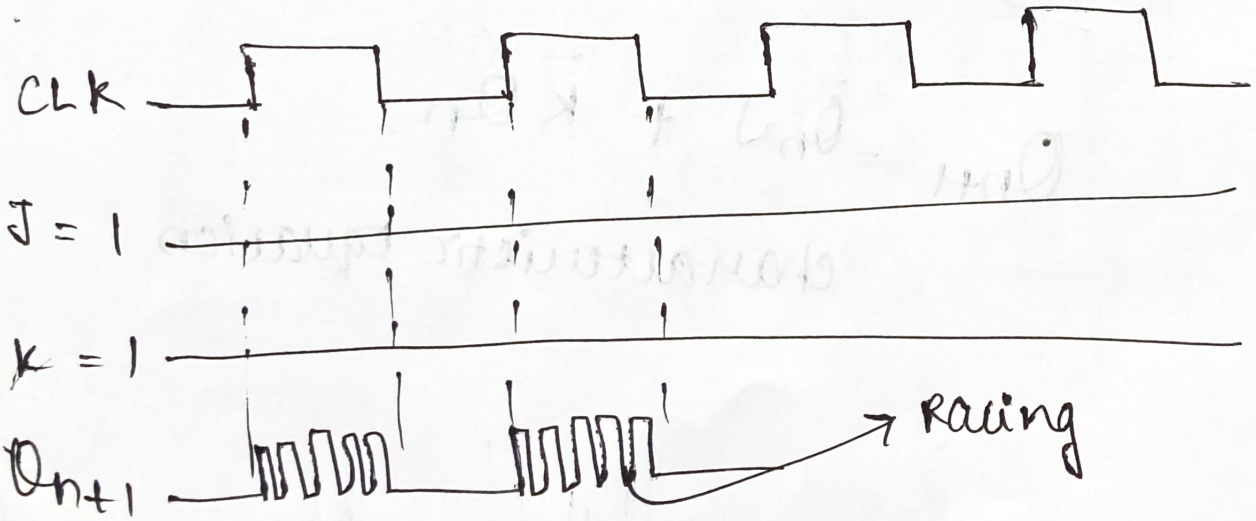
Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0





$J = Q_{n+1}$ $\quad K = \overline{Q_{n+1}}$

Toggle is controlled, but race around condition is not controlled.



Time required to produce output must be greater than $T/2$

conditions to overcome Racing

(i) $T/2$ < propogation delay of flip flop

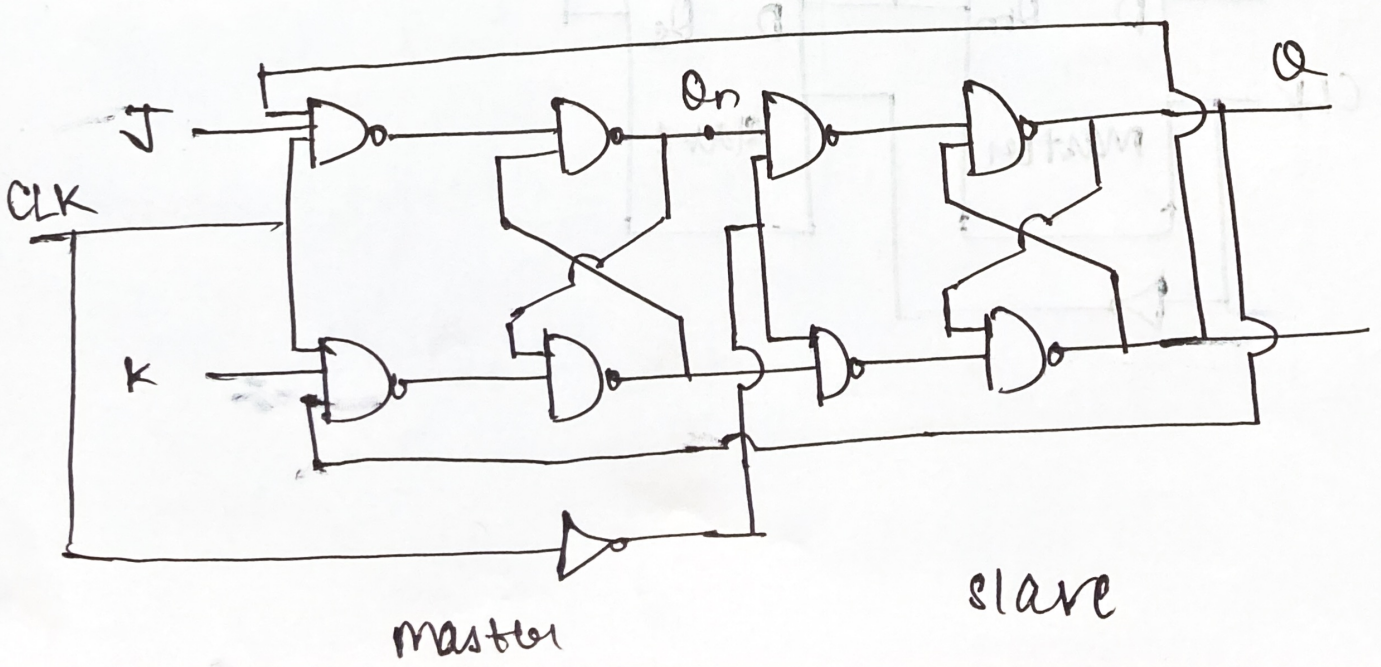
(ii) Using edge triggering.

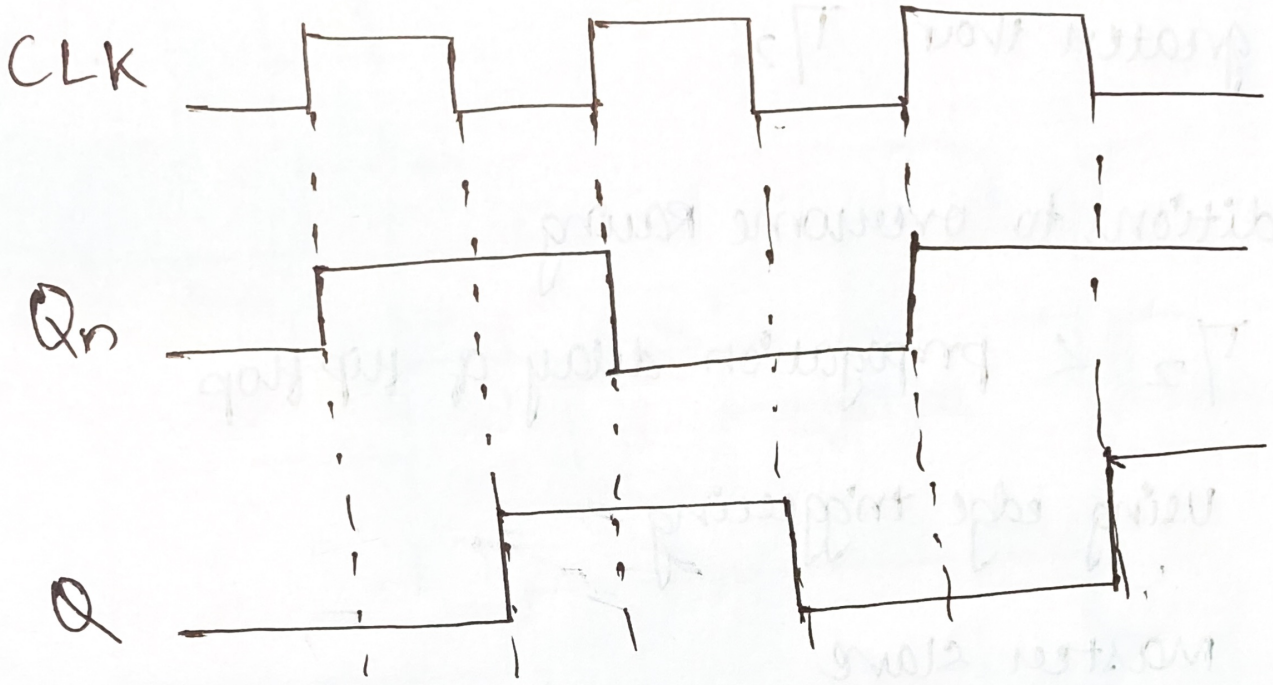
(iii) Master slave

$T/2$ (clock may time period)

Master slave operation is same as –ve edge triggered flip flop

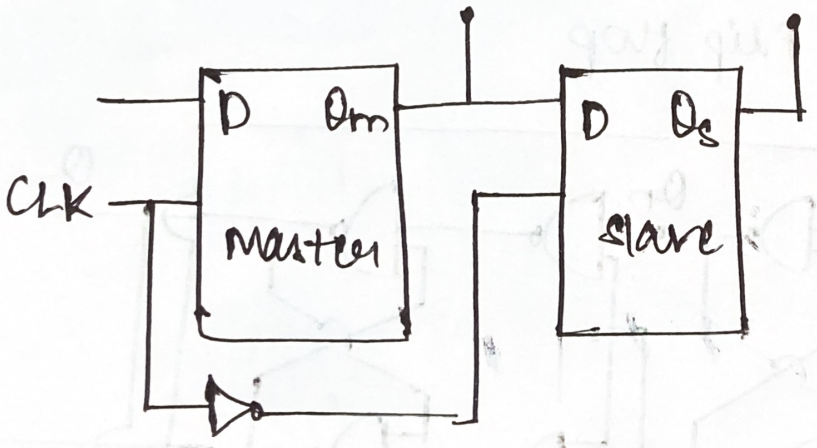
Master slave JK Flip flop

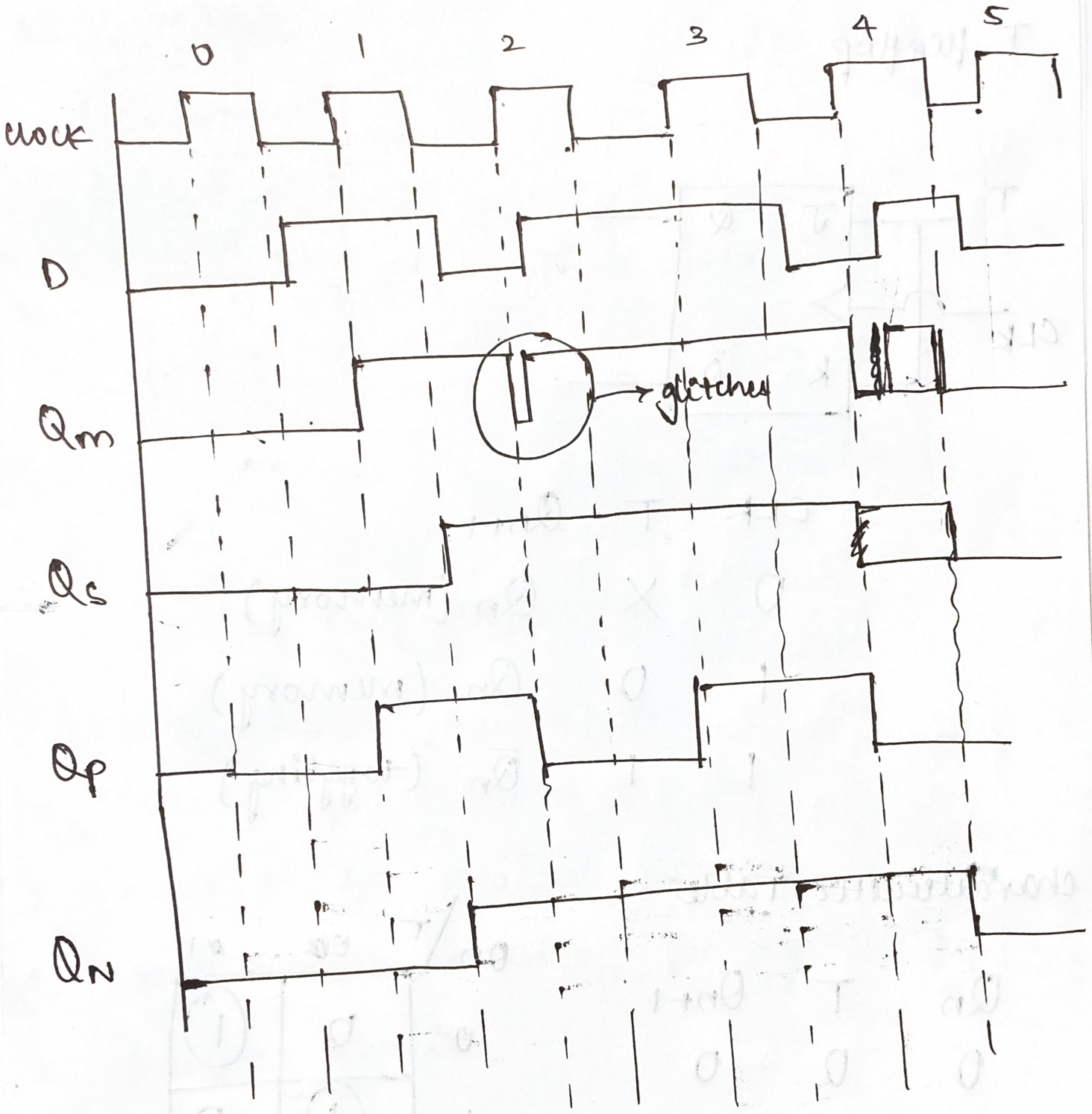




Output doesn't change continuously like in racing but it changes once in a cycle

Master slave D Flip Flop

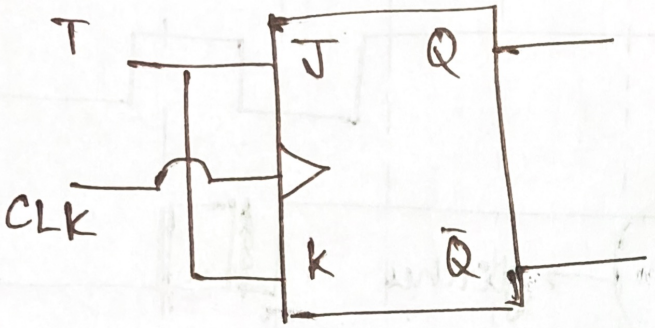




Master slave removes glitches

$$Q_S = Q_N$$

Ex:- T flip flop



CLK	T	Q_{n+1}
0	X	Q_n (memory)
1	0	Q_n (memory)
1	1	$\bar{Q}_n$ (toggling)

1.

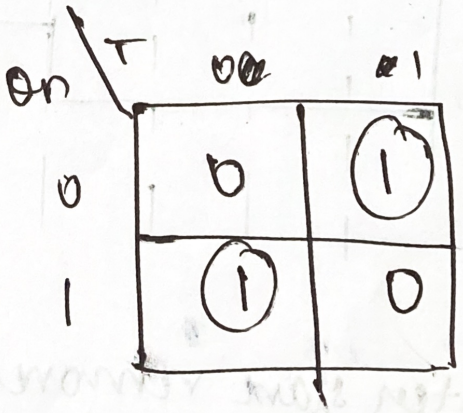
2.

3. Characteristic Table

4.

5.

Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0



$$Q_{n+1} = \bar{Q}_n T + Q_n \bar{T}$$

characteristic equation

$$Q_{n+1} = Q_n \oplus T$$

Excitation Table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

$$T = Q_n \overline{Q_{n+1}} + \overline{Q_n} Q_{n+1}$$

$$T = Q_n \oplus Q_{n+1}$$

Flip Flop Conversion

1. Identify available and required flip flop
2. Make CT for required flip flop
3. make ET for available flip flop
4. write boolean expression for available flip flop
5. Draw the circuit

JK TO D flip flop

available flip flop : JK

required flip flop : D

Q_n	D	Q_{n+1}	J	K
0	0	0	0	X
0	1	1	1	X
1	0	0	X	1
1	1	1	X	0

For J

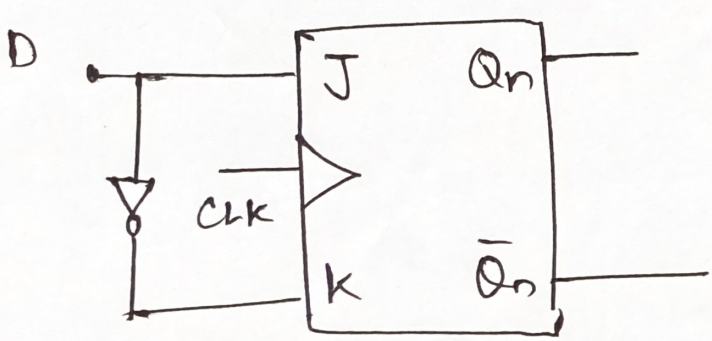
Q_n \ D	0	1
0	0	1
1	X	X

$J = D$

For K

Q_n \ D	0	1
0	X	X
1	1	0

$K = \bar{D}$



T to D flip flop

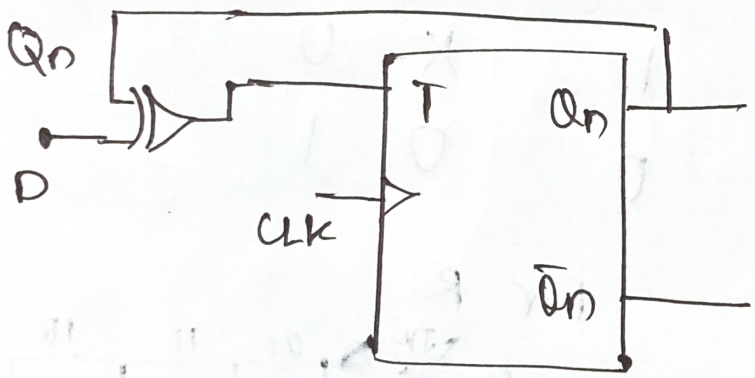
$Qw = T$

$req = D$

Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

Q_n	D	Q_{n+1}	T
0	0	0	0
0	1	1	1
1	0	0	1
1	1	1	0

$$T = Q_n \oplus D$$



SR TO JK flip flop

av :- SR

req = JK

Q_n	J	K	Q_{n+1}	S	R
0	0	0	0	0	X
0	0	1	0	0	X
0	0			1	0
0	1	0	1		
0	1	1	1	1	0
1	0	0	1	X	0
1	0	1	0	0	1
1	1	0	1	X	0
1	1	1	0	0	1

For S

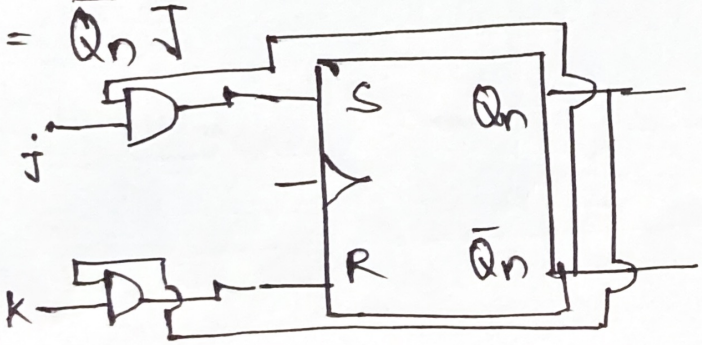
Q_n \ JK	00	01	11	10
0	0	0	1	1
1	X	0	0	X

$S = \bar{Q}_n J$

For R

Q_n \ JK	00	01	11	10
0	X	X	0	0
1	0	1	1	0

$R = Q_n K$



SR TO T flip flop

available = SR

required = T

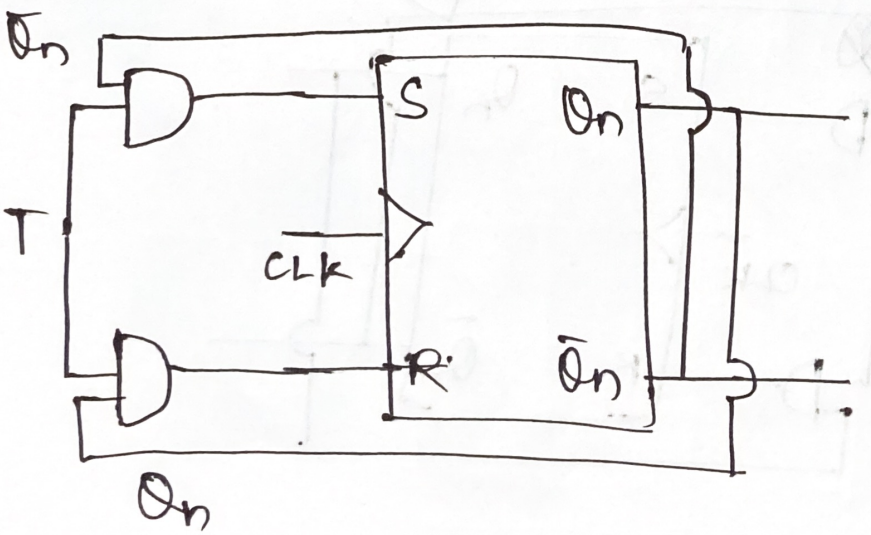
Q_n	T	Q_{n+1}	S	R
0	0	0	0	X
0	1	1	1	0
1	0	1	X	0
1	1	0	0	1

Q_n \ T	0	1
0	0	(1)
1	X	0

Q_n \ T	0	1
0	X	0
1	0	(1)

$S = \bar{Q}_n T$

$R = Q_n T$



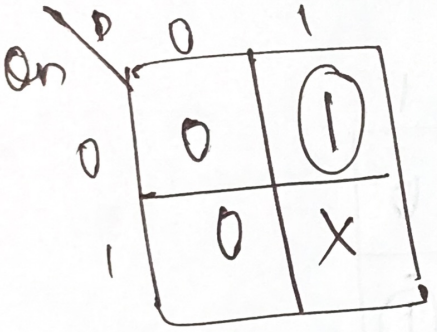
JK TO D

SR to D flip flop

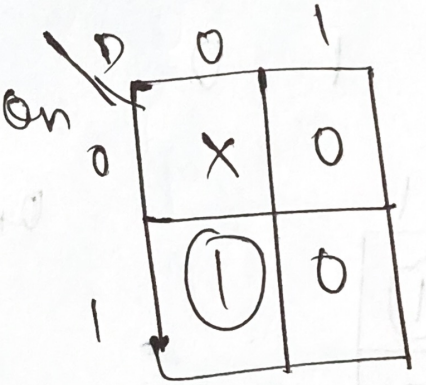
eqn = SR

req = D

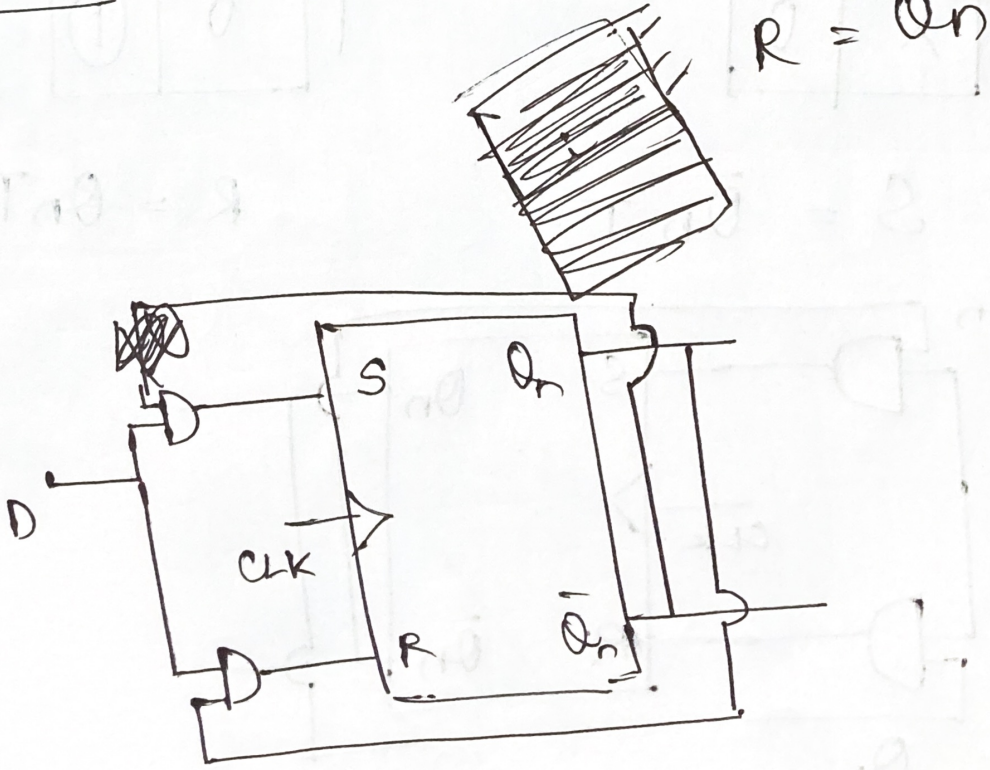
Q_n	D	Q_{n+1}	S	R
0	0	0	0	X
0	1	1	1	0
1	0	0	0	1
1	1	1	X	0



$S = \bar{Q}_n D$



$R = Q_n \bar{D}$



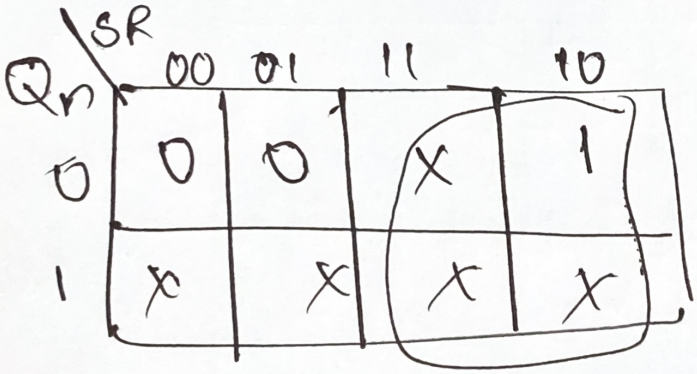
JK TO SR

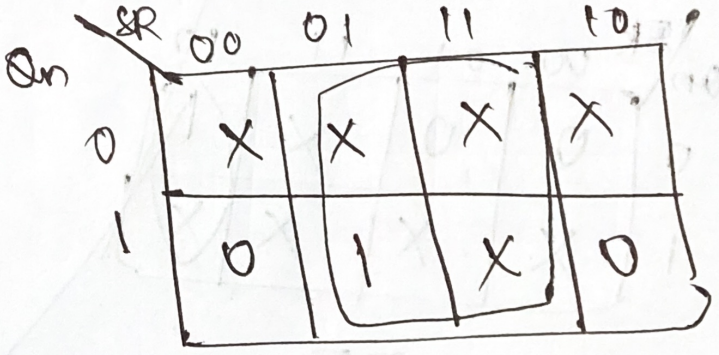
J

av = JK

nq = SR

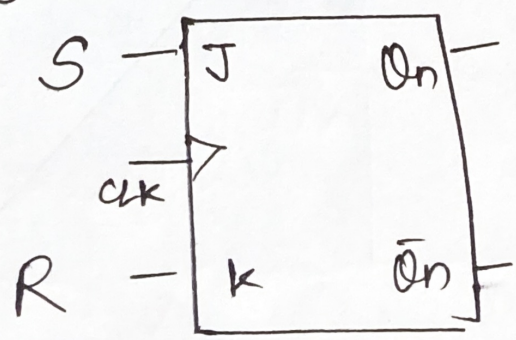
Q_n	S	R	Q_{n+1}	J	K
0	0	0	0	0	X
0	0	1	0	0	X
0	1	0	1	1	X
0	1	1	X	X	X
1	0	0	1	X	0
1	0	1	0	X	1
1	1	0	1	X	0
1	1	1	X	X	X





J = S

K = R



JK TO T

avl = JK

req = T

Q_n	J	K	Q_{n+1}	T
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	1	1
1	0	0	1	0
1	0	1	0	1
1	1	0	1	0
1	1	1	0	1

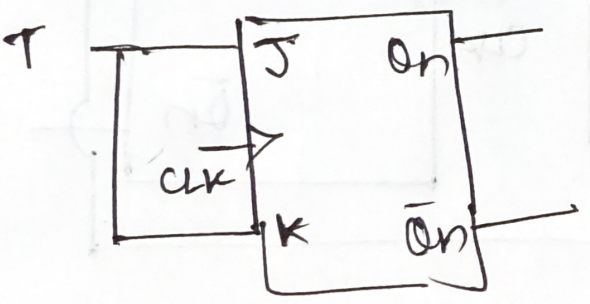
Q_n	T	Q_{n+1}	J	K
0	0	0	0	X
0	1	1	1	X
1	0	1	X	0
1	1	0	X	1

Q_n \ T	0	1
0	0	1
1	X	X

J = T

Q_n \ T	0	1
0	X	X
1	0	1

K = T



D

D TO SR

a

r $av = D$

$req = SR$

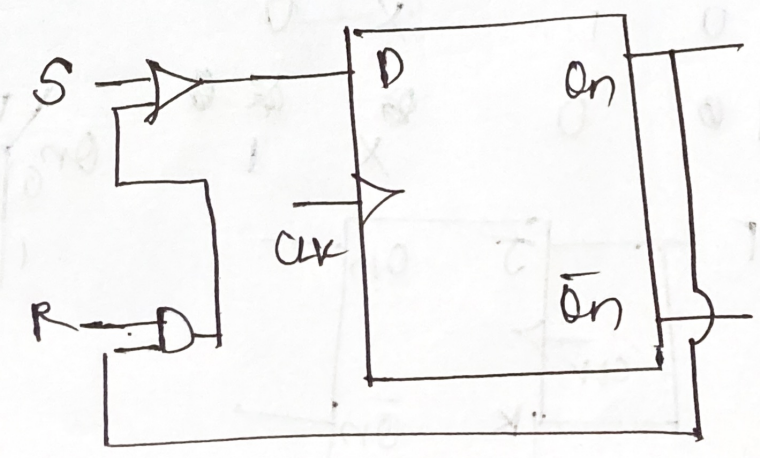
0

1

Qn	S	R	Qn+1	D
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	X	X
1	0	0	1	1
1	0	1	0	0
1	1	0	1	1
1	1	1	X	X

Qn \ SR	00	01	11	10
0	0	0	1	1
1	1	0	X	1

$$D = S + Q_n\bar{R}$$

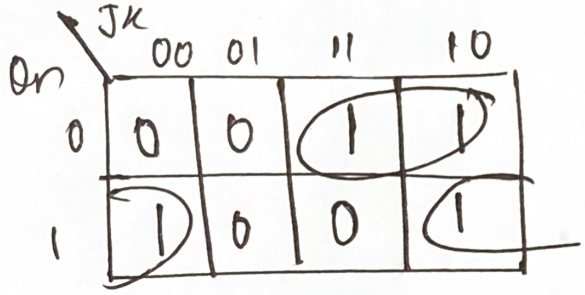


D TO JK

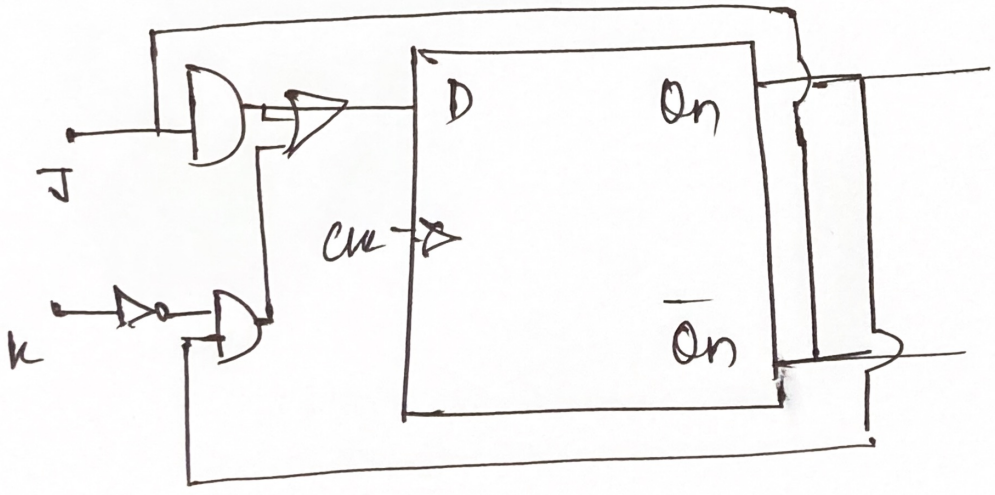
$Q_N = D$

$Req = JK$

Q_n	J	K	Q_{n+1}	D
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	1	1
1	0	0	1	1
1	0	1	0	0
1	1	0	1	1
1	1	1	0	0



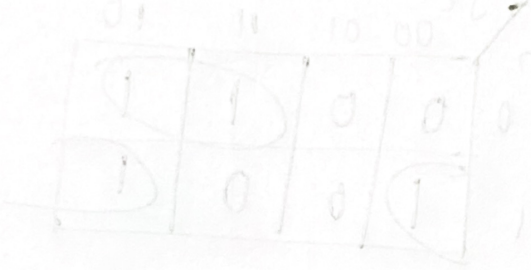
$D = J\bar{Q}_n + \bar{K}Q_n$

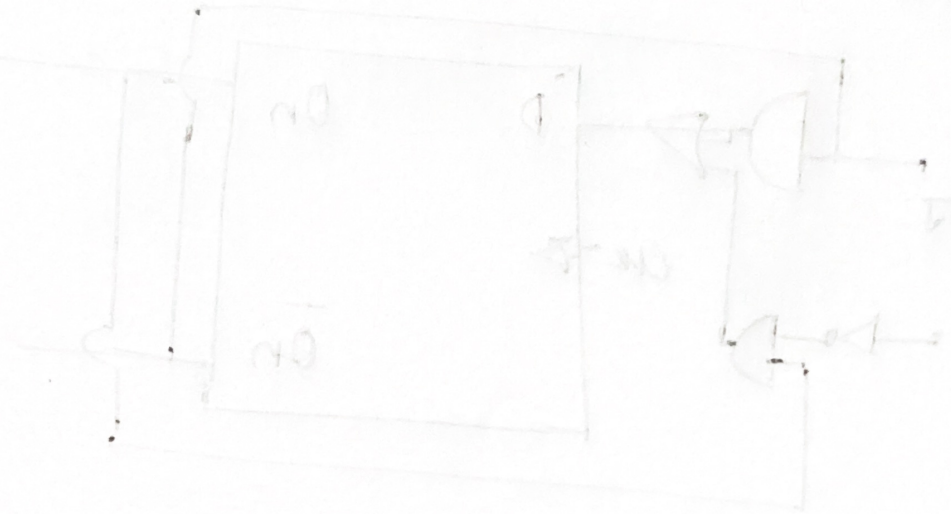


D TO T

$av = D$

$req = T$





T TO SR

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